

Subject code and title: MLN601 Machine Learning

Subject length: 6 and 12 Weeks

Subject Description:

This subject is designed to provide a graduate-level student an in-depth understanding of the methodologies, technologies, and algorithms currently used in machine learning. Students will be introduced to the theory behind a range of machine learning tools and practice applying the tools to different applications. It explores topics such as classification, linear models, supervised and unsupervised learning in conjunction with applied and theoretical machine learning concepts. Students complete practical activities using machine learning algorithms and methods.

Subject Learning Outcomes (SLO):

SLO a)	Evaluate and compare the key concepts in machine learning.
SLO b)	Apply learning algorithms to perform machine learning tasks.
SLO c)	Implement practical machine learning: data pre-processing, analysis, model selection, and interpret the results.
SLO d)	Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders.

Delivery Schedule:

	Module	Learning Activities	Assessment Progression & Due Date	SLOs addressed
1	Career Opportunities, Key Concepts, Applications, Types of Algorithms and Software	- Self-introduction - Hands on activities - Discussion forums	Familiarise yourself with the assessment briefs, coding, tools and technical instruction for hands-on with Notebooks.	a) d)
2	Managing Machine Learning Projects: CRISP-DM, Ethics by Design (Australasia) and Data Sets	- Hands on activities - Discussion forums	Attempt learning activities and familiarise yourself with the CRISP-DM template.	a) d)
3	Supervised Learning and Linear Regression	- Hands on activities - Discussion forums	Attempt learning activities and be familiar with the CRISP-DM template.	a) b) d)
4	Decision trees	- Hands on activities - Discussion forums	Attempt learning activities, review and submit Assessment 1. Assessment 1 due – Regression Analysis 20% Outcome: Notebook with mark-up (Source code and 1000 words (+/-10%))	a) b) c) d)
5	Classification and Bayes Rule	- Hands on activities - Discussion forums	Familiarise yourself with Assessments 2 and 3, and attempt learning activities.	a) b) c) d)
6	Support Vector Machines	- Hands on activities - Discussion forums	Attempt learning activities which are relevant for the preparation of Assessments 2 and 3.	b) d)
7	Automated and Explainable Machine Learning	- Hands on activities - Discussion forums	Attempt learning activities which are relevant for the preparation of Assessments 2 and 3.	a) d)

8	Logistic Regression	- Hands on activities - Discussion forums	Attempt learning activities, review and submit Assessment 2. Assessment 2 due: Classification 40% Outcome: Notebook with mark-p and presentation (7-10 min presentation, and source code, 1500 words (+/-10%))	a) b) c) d)
9	K-means Clustering	- Hands on activities - Discussion forums	Attempt learning activities which are relevant for the preparation of Assessments 3.	b) c) d)
10	Learning Theory: PAC	Discussion forum	Attempt learning activities which are relevant for the preparation of Assessments 3.	a) d)
11	The perceptron	- Hands on activities - Discussion forums	Attempt learning activities which are relevant for the preparation of Assessments 3.	a) d)
12	Future of Machine Learning (ML): Enterprise Grade ML and Weak Supervision	- Hands on activities - Discussion forums	Review and submit Assessment 3. Assessment 3 due: Machine learning project 40% Outcome: Notebook and model selection (up to 2000 words)	a) d)

Time Management:

Below outline the minimum time management expectations for students.

6 Week Delivery:

- 10 Hours per module: Facilitated study: 3 hours / module. Personal Study: 7 hours / module. (Typically 2 modules per week).
- 6 hours facilitated study consists of attending class, responding to facilitator feedback.

- Students are to allocate 14 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

12 Week Delivery:

- 10 Hours per module (one week): Facilitated study: 3 hours / week. Personal Study: 7 hours / week.
- 3 hours facilitated study consists of attending class, responding to facilitator feedback.
- Students are to allocate 7 hours of personal learning. This includes essential time spent on pre-reading and viewing materials, assessment progression and learning activities.

Learning Activities:

There are learning activities in each module with the types of activities listed in the delivery schedule above. These activities are an essential component of the subject and they are designed to prepare you for your graded assessment tasks. These learning activities will help you to build your understanding and skills whilst also collaborating with your peers.

***Detailed information of each activity can be found in their corresponding Module page in Blackboard.**